Week Plan, Oct 7–14: Sensing-Driven Meal Guidance

Oct 7, 2026 · @Sneha

This week at a glance

By end of day Wednesday, Oct 14, the team should have data access requested by all three members, a working repo, LH-anchored phase labels, and daily CGM metrics. Today, Oct 7, is the last day of proposal Week 1, so this plan covers today's close-out plus all of Week 2 (Oct 8–14).

The split below follows the proposal's three steps so each person owns one track through the semester. Nothing in the proposal assigns roles, so swap names freely if skills or interest point elsewhere.

Person	Track	Due Oct 14
Ava	Data access, cleaning, and phase alignment (Step 1)	Clean per-day table with a phase label for every participant-day
Madhav	CGM metrics and forecasting scaffold (Step 2)	Daily glucose metrics, synthetic dataset, and a baseline that runs end to end
Sneha	Literature and meal-guidance groundwork (Step 3)	Final literature review, evidence-card template, first 15 recipes

First steps for everyone

All three of you should sign the mcPHASES data use agreement today, because it is the only access requirement the dataset page lists. The proposal calls the dataset credentialed access with CITI training, but PhysioNet lists it as Restricted Access, so access may take hours instead of days.

- [x] ~~Submit the proposal today if it is not in yet. Consider correcting the access tier and the CGM coverage first (see the table below).~~

- [x] ~~Each person: create a PhysioNet account with your university email and sign the data use agreement. Signing is per person.~~
- [x] ~~Each person: download your own copy. The agreement forbids sharing access, so no raw files in the repo, a shared drive, or chat.~~
- [x] ~~If the logged-in page asks for anything beyond the agreement, do it the same day and tell the group.~~
- [ ] Madhav: create one private GitHub repo with `data/` in `.gitignore` and a pinned Python environment.
- [ ] Hold a 30-minute kickoff on Thursday, Oct 8 to settle the decisions listed at the end of this doc.

What the dataset page changes

Five details on the dataset page differ from what the proposal assumes. Each one affects this week's work.

Proposal assumes	Dataset page says	What to do this week
Credentialed access, CITI training, slow approval	Restricted access: registered users sign the agreement	Work on real data this week; keep synthetic data for the public demo
CGM across both 3-month intervals	CGM worn in Interval 1 only (2022)	Limit all glucose work (Step 1, target b, model 3) to Interval 1
42 participants in both intervals	42 in Interval 1; 20 returned for Interval 2 (2024), diaries optional	Report sample size per interval; decide if symptom models use Interval 2
Four phases you assign: menstrual, follicular, fertile, luteal	Ships Mira-derived labels: menstruation, late-follicular, ovulation, luteal	Pick the primary scheme; compare yours against the shipped labels
Real participant days go into the LLM in Step 3	Agreement bars sharing access; PhysioNet's LLM guidance recommends local models and requires zero data retention for APIs	Decide the LLM setup before any real data reaches an API

The LLM guidance is written about MIMIC but addresses credentialed and restricted PhysioNet data generally. Two more facts shape cleaning: dates are relative `day_in_study` values with Interval 2 resuming near day 905, and the authors applied no cleaning or imputation.

Ava: data, cleaning, and phase alignment

Ava owns the path from 23 raw tables to one clean per-day table with a phase label. Madhav's models and Sneha's evaluation both read from that table, so the data dictionary comes first.

- [ ] **Wed–Thu, data dictionary.** For each of the 23 tables, record grain, key columns, row count, participants covered, and interval. Save it as `docs/data_dictionary.md`.
- [ ] **Thu, confirm the target symptoms.** Check that fatigue, cramps, bloating, and cravings exist in the self-report table and note each scale. Raise any missing one at kickoff.
- [ ] **Fri, loader and interval split.** One load function per table, keyed on `id` and `day_in_study`. Tag every row as Interval 1 or Interval 2.
- [ ] **Fri, coverage report.** Per participant, count days with diary entries, hormone readings, Fitbit wear, and CGM. This sets the per-day coverage rules the proposal promises.
- [ ] **Mon, per-day table.** One row per participant-day with symptoms, LH, E3G, PdG, menstruation status, and the shipped phase label.
- [ ] **Mon–Tue, LH-anchored phases.** Find the LH surge in each cycle, then assign menstrual, follicular, fertile, and luteal. Flag cycles with no clear surge for separate analysis.
- [ ] **Tue, compare labels.** Cross-tabulate your phases against the shipped Mira labels and note where they disagree.
- [ ] **Wed, hand off.** Share the table schema and a one-page note on phase rules and exclusions.

Done when Madhav can join his daily glucose metrics to this table on `id` and `day_in_study` with no manual fixes. If time allows, start the calendar-estimated phase from period start dates alone, which the Week 4 models need.

Madhav: CGM metrics and forecasting scaffold

Madhav owns daily glucose metrics and the evaluation harness every Step 2 model will run through. He starts on synthetic data and switches to real data when Ava's per-day table lands on Monday.

- [ ] **Wed, repo and environment.** Private repo, `data/` ignored, folders for `src`, `notebooks`, and `docs`. Pin pandas, scikit-learn, one gradient-boosting library, statsmodels, and shap.
- [ ] **Thu, synthetic dataset.** Generate fake participants that match Ava's data dictionary. It unblocks the harness now and becomes the public demo data in Week 9.
- [ ] **Fri, daily CGM metrics (Interval 1 only).** Per participant-day: mean, median, SD, CV, MAGE, and time in range, using the consensus definitions the proposal cites. Store readings per day alongside.
- [ ] **Fri, coverage rule.** Dexcom G6 logs every 5 minutes, so a full day has 288 readings. Propose a minimum per day and apply it after Ava's coverage report.
- [ ] **Mon, glucose target.** Label high-variability days as CV in the top third of each participant's own days. Check label counts per person.
- [ ] **Mon–Tue, evaluation harness.** Leave-one-subject-out splits, AUROC and AUPRC with bootstrap confidence intervals, plus the persistence and per-person base-rate baselines.
- [ ] **Tue, first Fitbit daily features.** Resting heart rate, sleep duration, sleep efficiency, steps, and wear time. Other sensors wait for Week 3.
- [ ] **Wed, baselines on real data.** Run both baselines end to end for the glucose target and one symptom.

Done when one command prints AUROC and AUPRC for both baselines on real Interval 1 data.

Three suggestions to settle while building, since they are cheap now and costly later:

- A 70% per-day floor is a reasonable first coverage rule. Check it against the consensus paper before fixing it.
- Healthy participants will sit near 100% in the 70–180 mg/dL range, so also report a tighter 70–140 mg/dL range.
- Baselines and rolling features for a held-out participant must use only that person's past days, or the forecast leaks future data.

Sneha: literature and meal-guidance groundwork

Sneha closes the Week 1 literature review and builds the evidence base Step 3 depends on. Starting the recipe set now pulls Week 3 work forward and protects the tight LLM schedule in Weeks 7 and 8.

- [ ] **Wed–Thu, finalize the literature review.** Add the dataset authors' 2024 wrist skin temperature paper in npj Women's Health, which bears on the temperature feature. Confirm every reference resolves by DOI or URL.
- [ ] **Thu, check reference 2.** The dataset page lists that paper's first author as G. Lin with four co-authors before Truong and Mariakakis. Match the citation to the journal page.
- [ ] **Thu, LLM data-handling options.** One page comparing a local model, an API with zero data retention, and sending only synthetic or coarse derived state. Bring a recommendation to kickoff.
- [ ] **Fri, evidence table.** For each of the four symptoms, glucose-aware planning, and the iron-during-menses fallback: the dietary claim, best source, study type, and strength. Mark claims with no good source.
- [ ] **Mon, evidence-card template and first 5 cards.** Fields: claim, food or dose, population studied, source, strength, cautions.
- [ ] **Mon, recipe schema.** Ingredients in grams with FoodData Central IDs, dietary pattern tags, allergens, cooking time, and meal type.
- [ ] **Tue, first 15 of about 50 recipes.** Spread them across omnivore, vegetarian, and vegan.
- [ ] **Tue–Wed, nutrition check.** Get a FoodData Central API key and compute nutrients in code for 3 recipes.
- [ ] **Wed, share the evidence table.** Flag any symptom where the evidence is weak, so guidance for it can be labeled or dropped.

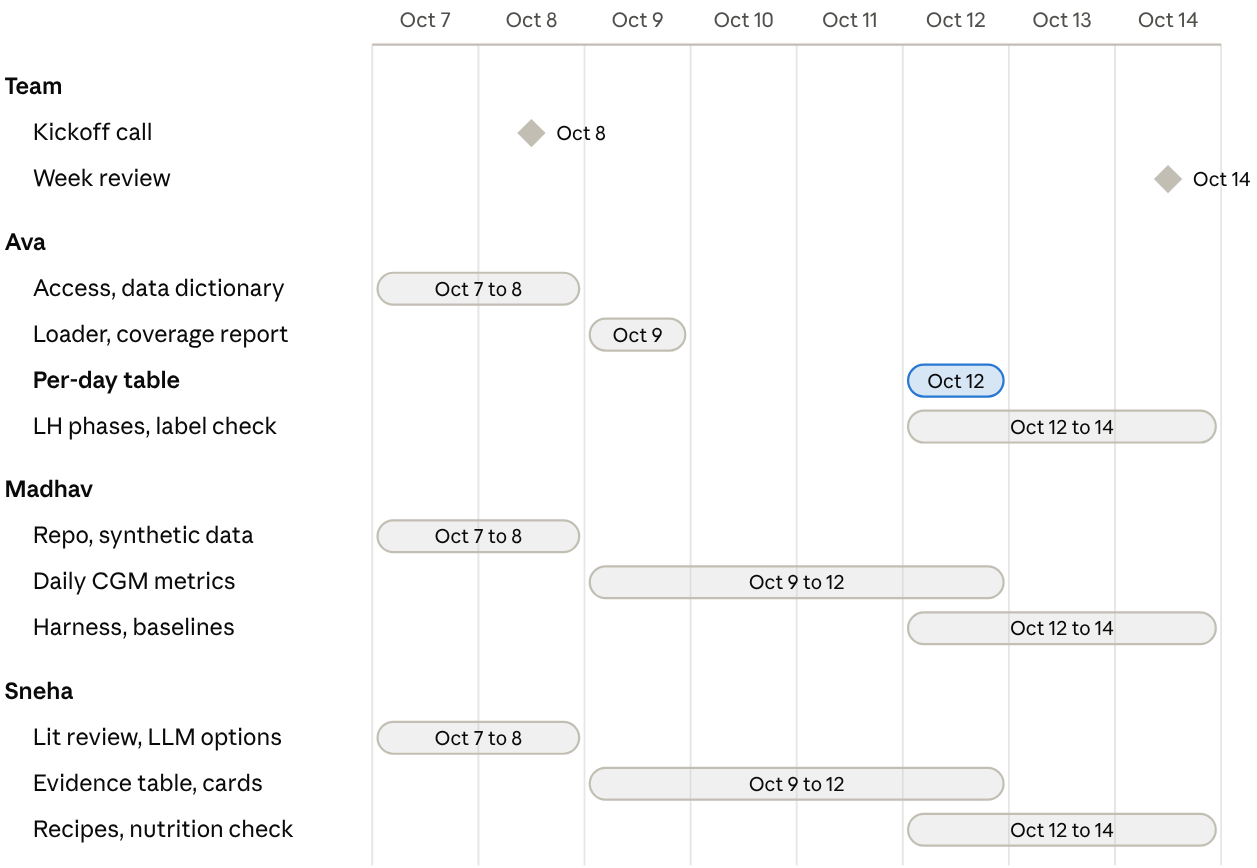
Done when every planned guidance rule traces to a source or is marked unsupported, and nutrition for 3 recipes computes from code. Sneha's tasks this week do not need the dataset, but she should still sign the agreement now for the Step 3 evaluation.

Schedule and check-ins

The three tracks run in parallel until Monday, Oct 12, when Ava's per-day table lets Madhav move from synthetic to real data.

Real-data baselines wait on the per-day table

Week plan by day, 2026



week timeline · 3 tracks, 2 team check-ins

Bars spanning Oct 9 to 12 cross the weekend; they mark the window, not weekend work. Meeting times are suggestions, so move them to fit your class schedules.

- **Thu, Oct 8, kickoff (30 min).** Confirm access for all three, settle the three decisions below, and review Ava's symptom check.
- **Mon, Oct 12, async update.** Each person posts what is done and what is blocked. Ava shares the per-day table schema.
- **Wed, Oct 14, week review (30 min).** Each person shows their deliverable, then the group plans the Step 1 replication for Week 3.

Decisions and risks

Three decisions need a group answer at Thursday's kickoff, because each one changes what gets built this week.

1. **Primary phase scheme.** Suggested: your own LH-anchored phases as primary, since the proposal commits to them, with the shipped Mira labels as a cross-check.
2. **Interval scope.** Glucose work must use Interval 1. Suggested: make Interval 1 primary for symptoms too, and add Interval 2 as a sensitivity analysis if Ava's coverage report shows enough diary days.
3. **LLM setup for Step 3.** Suggested: pick from Sneha's options page, then confirm with the instructor before any real participant day reaches a hosted model.

Risk this week	How you will know	Fallback
A target symptom is not in the diary	Ava's check on Thursday	Use the nearest recorded symptom and note the change for the report
One member's access is delayed	No file access by Thursday evening	That member works on synthetic data; nobody shares a download
Too few usable CGM days after the coverage rule	Coverage report on Friday	Loosen the threshold or report on fewer participants
Madhav is blocked on Ava's table	No schema by Monday	Build against the data dictionary and synthetic data
Proposal wording is out of date	Access tier and CGM coverage differ from the dataset page	Correct before submitting, or note both in the final report

Sources

- mcPHASES v1.0.0 dataset page, PhysioNet: access tier, intervals, devices, phase labels, table structure
- PhysioNet Restricted Health Data Use Agreement 1.5.0: terms on sharing access and code release
- Use of MIMIC Data with Large Language Models and Online Services, PhysioNet, Sept. 24, 2025
- The team's project proposal, COMS 6998, Fall 2026: research questions, steps, and timetable